

The Fundamentals of Programming

Variables & Assignment Operators



Objectives

What you will learn:

1. How to declare variables using the assignment operator and assign them values.
2. The many different assignment operators available in Python.
3. What a “type” is and why it’s important.

Variables

Back to math class

At some point you've probably seen a problem like the one to the right in your math class. The ever present **x** variable, right?

What you should remember is that a variable is a placeholder for a potential value.

Billie went to the movies. She spent \$20. If her ticket was \$14, how much was her drink?

a. $20 + x = 14$

b. $20 \div x = 14$

c. $14 + 20 = x$

d. $14 + x = 20$

Variables

The difference

Variables in programming work similar to those in math class, but often look quite different. With the variables on the right, notice how they mostly use descriptive names and not single letters. They also come in different **types**, some containing numbers and others containing words.

```
age = 35
temperature = 31.7
name = 'Aiko'
cash = 1050
address = '115 Market St.'
filename = "myhomework.doc"
i = 0
x = 20
```

What else do you notice?

Declaring variables

The Rules

When creating a variable you **declare** it. When declaring variables in Python there are some basic rules you must following.

Legal variables

time	counter1
x	monkey_1
first_name	this_var25
_lastname	_a_

Illegal variables

counter1!	@variable
50temp	my variable
1234	variable 1
#num	number-25

Types

The basics

Whenever you create a variable Python will assign it a **type** based on what you put on the right side of the equal sign. The four basic types are: **integer**, **float**, **string** and **Boolean**.

Float

```
money = 56.50  
temp = 14.1  
dec = 0.002
```

String

```
name = "Ted"  
gender = "M"  
grade = "A+"
```

Integer

```
age = 27  
total = 50004
```

Boolean

```
is_weekend = True  
temp = False
```

The Assignment Operator

This is often a point of confusion for those new to programming, but there is one operator that doesn't entirely work the same as in your math class, and that is the equal sign.

Look at these examples:

```
age = 10  
greeting = 'Hello'  
is_tired = True
```

In math class this is read as:
“age is equal to 10”

In Python this is read as:
“age is assigned the value of 10”

As we'll see, this is a big difference.

Calculations

The subtle difference

Math can do:

$a = b + 1$

Math can't do:

$a = a + 1$

That last equation makes no sense as the left is not equal to the right.

Python can do:

$a = b + 1$

Python CAN do:

$a = a + 1$

The last statement is not an equation, but an assignment, saying we're replacing **a** with **$a + 1$**

Calculations

A bit of practice

Given the variables below, and the assignment statements to the right, find the final value of each of the variables.

a = 10
b = 14
c = 1
d = 1.5

- 1. $a = a * 5$**
- 2. $b = b + c$**
- 3. $d = (d * 2) + a$**
- 4. $a = \text{abs}(a - 15)$**
- 5. $c = \text{float}(\text{pow}(c+3, 2))$**
- 6. $c = \text{min}(c, a) * \text{max}(b, a) + 1$**

Pay attention to the type!

Other Assignment Operators

In Python there is more than one assignment operator, though most are just shorthand for doing things quickly.

Use the following variables:

`money = 1000`

`count = 10`

`age = 17`

See if you can figure out the final value for these assignments:

1. `money += 1000`
2. `count -= 1`
3. `age *= 2`
4. `count %= 4`
5. `money /= count`

Practice

Putting it together

What will be the final output from these two programs? Pay close attention to the types involved.

```
students = 30
days = 5
total_meals = students * days
print("Meals:", total_meals)
```

```
x = max(5, 4, 3, 7.0)
y = min(1, 2, 3 % 3)
z = 2
z += z * (x + y)
print(x, y, z)
```

Recap

We learned

1. How to assign variables and determine their type.
2. How the `=` operator differs from math to Python.
3. How to use a few different operators that work as shortcuts for doing assignments and arithmetic operations.